

Qian Yang

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🎓 EDUCATION

Mila - Quebec AI Institute & Université de Montréal Ph.D. in Computer Science CGPA: 4.3/4.3 Advisor: Prof. Aishwarya Agrawal Research: Efficiency and Reliability of Multi-modal Large Language Models	2023.09 – Present
Harbin Institute of Technology, Shenzhen MSc in Computer Science and Technology Advisor: Prof. 户保田教授 Research: Fine-grained Alignment for Explainable Multi-modal Inference	2020.09 – 2023.03
University of Electronic Science and Technology of China BEng in Computer Science and Technology CGPA: 3.73/4.0 (Top 10%)	2016.09 – 2020.06

⚡ RESEARCH HIGHLIGHTS & CORE COMPETENCIES

- **Efficient Multimodal Training:** Focused on **Data-centric AI** for VLMs. Developed the SAIL framework (CVPR'25 Highlight) achieving CLIP-level performance with **6% data**, and the PROGRESS framework (CVPR'26) for high-efficiency instruction tuning.
- **VLM Reliability & Factuality:** Specialized in quantifying and mitigating hallucinations in Large Models. Developed task-decomposition consistency metrics (EMNLP'24) and execution-based code hallucination benchmarks (AAAI'25).
- **Distributed RL Systems:** Proficient in large-scale Reinforcement Learning using **verl** and **DeepSpeed**. Experienced in designing Fine-grained Reward Models and optimizing training efficiency in multi-node GPU clusters.

🏢 PUBLICATIONS

- **Qian Yang***, Shivam Chandhok*, ..., Leonid Sigal, Aishwarya Agrawal. Learning What Matters: Prioritized Concept Learning via Relative Error-driven Sample Selection. CVPR 2026.
– **Efficient Tuning:** A dynamic sample selection framework for VLM instruction tuning that **reduces data and compute costs by 80%** while maintaining SOTA performance across multimodal benchmarks.
- Le Zhang, **Qian Yang**, Aishwarya Agrawal. Assessing and Learning Alignment of Unimodal Vision and Language Models. CVPR 2025 (**Highlight**).
– **Low-Resource Alignment:** Introduced the SAIL framework, matching CLIP's zero-shot performance using **only 5 hours of single-GPU training** and 6% of the standard paired dataset. Demonstrated the extreme efficiency of unimodal-to-multimodal alignment.
- Yuchen Tian, Weixiang Yan, **Qian Yang**, ..., Dawn Song. CodeHalu: Investigating Code Hallucinations in LLMs via Execution-based Verification. AAAI 2025.
- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Decompose and Compare Consistency: Measuring VLMs' Answer Reliability via Task-Decomposition Consistency Comparison. EMNLP 2024.
- Le Zhang, Yihong Wu, **Qian Yang**, Jianyun Nie. Exploring the Best Practices of Query Expansion with Large Language Models. EMNLP Findings 2024.
- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Enhancing Multi-Agent Multi-Modal Collaboration with Fine-Grained Reward Modeling. *NeurIPS 2024 Workshop on AFM*.
- **Qian Yang**, ..., Min Zhang. Enhancing Multi-modal and Multi-hop Question Answering via Structured Knowledge and Unified Retrieval-Generation. ACM MM 2023.
- **Qian Yang**, Yunxin Li, ..., Min Zhang. Chunk-aware Alignment and Lexical Constraint for Visual Entailment with Natural Language Explanations. ACM MM 2022.
- Yunxin Li, **Qian Yang**, ..., Lin Ma. Fast and Robust Online Handwritten Chinese Character Recognition with Deep Spatial & Contextual Information Fusion Network. TMM 2022.

Preprints

- **Qian Yang**, ..., Aishwarya Agrawal. How and What to Imagine? Visual Thinking in Unified Multimodal Models for Cross-View Spatial Reasoning (Under Review) 2026
- Kanishk Jain, **Qian Yang**, ..., Aishwarya Agrawal. Discovering Failure Modes in Vision-Language Models using RL (Under Review) 2026

👜 ACADEMIC INTERNSHIPS

Tencent Hunyuan Team, China (<i>Qingyun Elite Internship Program</i>) Reward Modeling for Text-to-Image (T2I) Generation	2026.06 – Present
• Unified Reward Model for T2I: Training a reward model for text-to-image generation that fuses objective rubric-based criteria	

with **subjective human preference**, producing more reliable and interpretable alignment signals for image generation.

Alibaba DAMO Academy, Hangzhou, China

2022.05 – 2022.10

Multimodal Retrieval-Augmented Generation (RAG) & Knowledge Alignment | Advisors: Dr. Wen Wang, Dr. Qian Chen

- **Multimodal RAG Architecture:** Developed a unified **Retrieval-Generation** framework to integrate intermediate cross-modal reasoning steps. Effectively mitigated information gaps in multi-hop question answering, significantly enhancing the model's reasoning depth.
- **Entity-Centric Fusion:** Architected an entity-based alignment model utilizing structured knowledge graphs to bridge the semantic gap between visual features and textual concepts. Solved the "long-tail entity" recognition challenge.
- **Impact:** Implemented the pipeline within the team's research stack; findings published in **ACM Multimedia 2023**.

AWARDS AND SCHOLARSHIPS

FRQNT (Fonds de recherche du Québec), Québec, Canada (25,000 CAD/year)	2026 – 2029
AI Scholarship , University of Montreal (10,000 CAD)	2026
Women in Machine Learning (WiML) , MILA (7,000 CAD)	2025
Student Success Scholarship , University of Montreal (5,000 CAD)	2025
DIRO Excellence Scholarship , University of Montreal (5,000 CAD)	2024 – 2025
Professor Cho Diversity Scholarship , MILA (1,500 CAD)	2025
National Encouragement Scholarship (Top 10%, National Level Award)	2019
First Prize Academic Scholarship , UESTC (Top 20%)	2016 – 2020

ACADEMIC SERVICE & ACTIVITIES

- **Workshop Organizer:** **Vision Language Models For All (VLMs4All)**, *CVPR 2025 Workshop* – co-organizing a venue on Building Geo-Diverse and Culturally Aware VLMs.
- **Reviewing:** **CVPR 2025 Outstanding Reviewer**; Program Committee / Reviewer for CVPR 2024-26, ECCV 2024, AAAI 2024, ACM Multimedia 2023-24, COLING 2022.
- **Teaching:** *Representation Learning* (IFT6135, Fall 2024, Univ. of Montreal) – graduate-level TA (assignments, lab sessions, technical guidance); *Mathematical Logic / Intro to Algorithms* (2020-2021, HIT Shenzhen).

TECHNICAL SKILLS

- **Programming:** Python (Expert), C/C++, Shell Scripting (Bash)
- **DL & Distributed Systems:** PyTorch, verl (Ray + vLLM), DeepSpeed (ZeRO 1/2/3), FlashAttention
- **Infrastructure & Tools:** SLURM Cluster Management, Linux (Ubuntu/CentOS), Git, Docker, WandB
- **Languages:** English (TOEFL: 99 — R:26, L:25, S:22, W:26), Mandarin (Native)